

A Japanese Language Model and Three New Evaluation Benchmarks for Pharmaceutical NLP

JPharmatron & JPharmaBench

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1. Motivation: Why Pharmaceutical NLP?
2. JPharmaBench: Three Novel Benchmarks
3. JPharmatron: Building the Model
4. Results & Discussion

Why Pharmaceutical NLP Matters

The Challenge: LLMs still struggle in domain-specific settings where deep contextual understanding and precise terminology handling are required.

- **Japanese pharmaceutical industry** faces significant administrative overhead:
 - Document preparation & verification
 - Regulatory compliance (GMP, ICH guidelines)
 - Quality assurance workflows

Gap: No LLMs or evaluation benchmarks tailored for Japanese pharmaceutical operations!

- Medical domain $\neq$ Pharmaceutical domain
- Different terminology, regulations, and workflows

Our Contributions

1. JPharmatron

First Japanese LLM specialized for pharmaceutical operations

- Continual pre-training on **2B Japanese + 8B English** pharmaceutical tokens

2. JPharmaBench

First benchmark suite for Japanese pharmaceutical NLP

- Three benchmarks reflecting real-world tasks

Step1. Data Curation



Japanese Medical Articles
PubMed Abstracts



PMDA
Official documents



Swallow
CC100

Step2. Model Development

 **JPharmatron**

Continual Pretraining (+ Evolutional Model Merging)

Step3. Evaluation with Japanese pharma-related tasks



JPharmaBench

IgakuQA
JMMLU



YakugakuQA

factual recall, question-answering

NayoseQA

terminology normalization

SogoCheck

inconsistency detection

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JPharmaBench: Three Complementary Benchmarks

Benchmark	Format	Main Skill	Source	#Examples
YakugakuQA	5-choice QA	Factual recall	Licensing exams	3,021
NayoseQA	5-choice QA	Terminology normalization	KEGG DRUG	34,769
SogoCheck	Sentence pair	Inconsistency detection	Japanese Pharmacopoeia	200

Key Design Principle: Designed to reflect practical scenarios in pharmaceutical companies

- Covers **academic knowledge** (YakugakuQA)
- AND **real-world operational workflows** (NayoseQA, SogoCheck)

YakugakuQA: National Licensing Exam

Source: Japanese National License
Examination for Pharmacists (2012-2024)

3,021 questions across **9 categories**

Pharmacy

Pharmacology

Chemistry

Pathology

Hygiene

Physics

Practice

Law

Biology

Example Question

Which of the following is NOT an ideal property of a dilute solution?

1. Vapor pressure lowering
2. Freezing point depression
3. Boiling point elevation
- 4. Surface tension reduction ✓**
5. Osmotic pressure

NayoseQA: Terminology Normalization

Task: Resolve lexical variations of drug/chemical names

34,769 examples from KEGG DRUG Database

Why It Matters

- Essential for **drug interaction checks**
- **Medical record standardization**
- **Multilingual information retrieval**

Japanese name



English name

Brand name



Generic name

Chemical name



Common name

Example

"What is the English name for 常水?"

→ **Water**

SogoCheck: Cross-Document Inconsistency Detection

Task: Detect logical/factual inconsistencies between paired pharmaceutical texts

200 examples from Japanese Pharmacopoeia

"**Sogo**" = cross-document validation in pharmaceutical QA workflows

Example

Text A: Storage method: sealed container.
Temperature below 25°C . Humidity below 60% .

Text B: Storage method: sealed container.
Temperature below 26°C . Humidity below 61% .

Labels: Change in temperature Change in humidity

Key Insight: Even **GPT-4o** achieves only **39.1%** accuracy! Cross-sentence consistency reasoning remains an **open challenge**.

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Building JPharmatron

Goal: Not to beat GPT-4o, but to build a **practically deployable** model that balances accuracy, efficiency, and **privacy** for real-world use.

Base Model: Qwen2.5-7B

Multilingual support & Commercially permissible

Method: Continual Pre-Training (CPT)

Training Variants:

2B JP pharma

+8B EN PubMed

9B deduped ✓

Training Data Sources:

- Journal articles (pharmacology, pharmacy)
- PubMed abstracts (~8B tokens)
- PMDA package inserts
- Government documents
- General corpus (FineWeb, Swallow)

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Evaluation Results

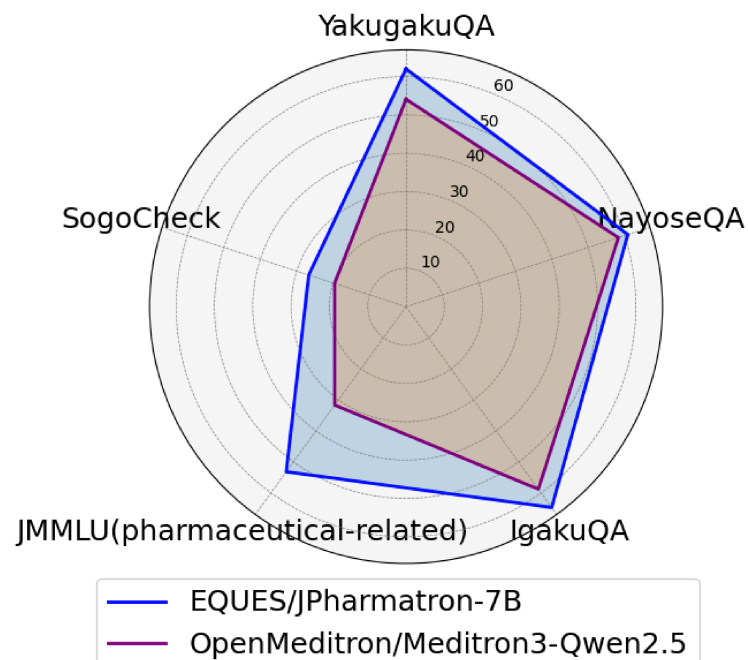
Model	YakugakuQA	SogoCheck	IgakuQA
Meditron3-Qwen2.5-7B	54.1	19.6	58.8
JPharmatron (best)	62.0	26.7	64.7
GPT-4o	83.6	39.1	86.6

Key Findings:

+7.9% over Meditron on YakugakuQA

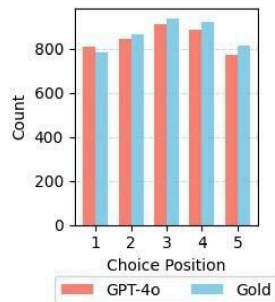
Medical specialization alone is **insufficient**

SogoCheck is **challenging for ALL models**

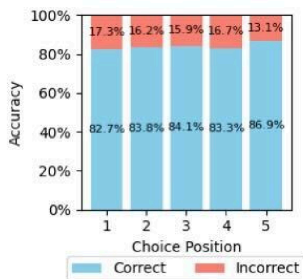


JPharmatron vs Meditron comparison

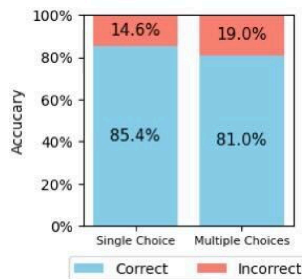
What We Learned: Error Analysis



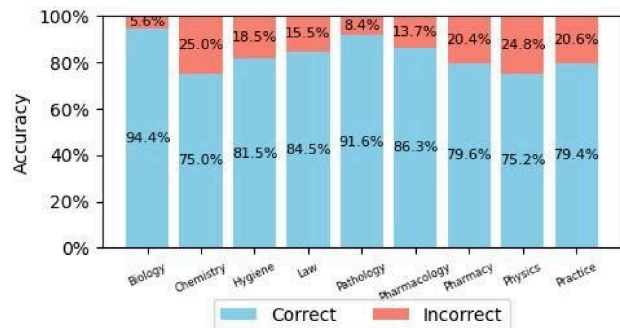
(a) Positional bias (count)



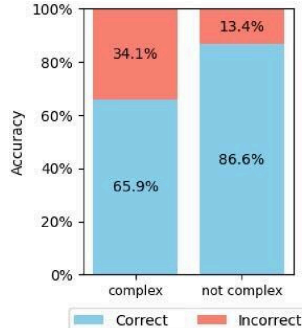
(b) Positional bias (error rate)



(c) Single-choice vs. Multiple-choice



(d) Category-wise accuracy



(e) Complex questions

Key Findings from GPT-4o Analysis:

Positional bias: Favors option "1"

Category performance:

Biology ~94% ✓

Chem ~75% ✗

Complex questions: 34.1% error rate

Takeaway: Top-tier LLMs still struggle with calculation-intensive pharmaceutical tasks.

Summary & Resources

- **First** Japanese pharmaceutical LLM and benchmark suite
- Demonstrates feasibility of **cost-effective, privacy-conscious** domain adaptation
- **SogoCheck** reveals open challenges in cross-document consistency reasoning

Resources:

Model & Data: huggingface.co/collections/EQUES/jpharmatron

Benchmarks: huggingface.co/collections/EQUES/jpharmabench

Thank you!